

Master's Thesis

Adaptive Rounding for Efficient Post-Training Quantization of Deep Neural Networks

48246347 Piye Liu

Supervisor Prof. Hiroshi Nakamura

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Department of Information Physics and Computing
Graduate School of Information Science and Technology
The University of Tokyo

概要

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Abstract

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Chapter 1

Introduction

Chapter 2

Background

Chapter 3

Literature Review

Chapter 4

Proposed Method

4.1 Proposed Method

To reconcile AdaRound’s assumption that the gradient term in ?? can be ignored, and Central Flows’ evidence that the gradient is nonzero even for pretrained converged models, we propose a modified version of AdaRound to incorporate awareness of the local gradient.

In order to minimize,

$$\mathbb{E} \left[g^{(w)^T} \Delta W \right], \quad (4.1)$$

we can attempt to force ΔW to have the same sign as $-g^{(w)^T}$. We approximate this by finetuning a model on a small amount of training data, and compute the Exponential Moving Average (EMA) of the model during each iteration. We then use $\text{sign}(W'_{\text{avg}} - W)$, the sign of the difference between the weights before and after finetuning, as a proxy to $-\text{sign}(g^{(w)^T})$. Finally, we add the Binary Cross Entropy between the current rounding directions and $\text{sign}(W'_{\text{avg}} - W)$ as a penalty term to AdaRound’s optimization objective.

Algorithm 1 Proposed Method (Finetuning)

Require: W , X (calibration data), η (learning rate), B (batch size)

Ensure: Finetuned model W' , EMA of models during finetuning W'_{avg}

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1:  $W' \leftarrow W$ 
2:  $W'_{\text{avg}} \leftarrow W$ 
3:  $N \leftarrow \frac{X}{B}$  ▷ Number of Iterations
4:  $\alpha \leftarrow \frac{2}{N+1}$  ▷ EMA Smoothing Factor
5: for each batch  $x_b$  in  $X$  do
6:    $\hat{x}_b \leftarrow W'(x_b)$  ▷ Forward pass
7:    $\mathcal{L} \leftarrow \text{CrossEntropyLoss}(\hat{x}_b, x_b)$ 
8:    $W' \leftarrow W' - \eta \cdot \nabla \mathcal{L}$  ▷ Backward pass
9:    $W'_{\text{avg}} \leftarrow \alpha \cdot W'_{\text{avg}} + (1 - \alpha) \cdot W'$ 
10: end for
11: return  $W'$ ,  $W'_{\text{avg}}$ 
```

Algorithm 2 Proposed Method (Calibration)

Require: W , W'_{avg} , X (calibration data), η (learning rate), t (number of iterations)

Ensure: Quantized model W_Q TODO

```

1: return  $W_Q$ 
```

Chapter 5

Evaluation

Chapter 6

Conclusion

Bibliography

A

Appendix